Linear Functions: Forms, Intuition, and Inverses

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Motivation

Linear functions are the simplest yet most fundamental class of functions.

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Linear functions are the simplest yet most fundamental class of functions.

- They describe **uniform change** and **straight lines**.
- They appear everywhere: physics, economics, and geometry.
- They serve as **building blocks** of linear transformations.

Definition

Linear Function

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$$f(x) = mx + c, \qquad m, c \in \mathbb{R}.$$

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- m: slope — rate of change.
- c: intercept — value of f when $x = 0$.

Different Forms of Linear Equations

Form	Equation	Use
Slope–Intercept	$y = mx + c$	Quick graphing
Point–Slope	$y - y_1 = m(x - x_1)$	Given one point
Two–Point	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$	Two points known
General	$Ax + By + C = 0$	Algebraic relations
Vector	$\vec{r} = \vec{r_0} + t\vec{v}$	Geometry / vector form

Geometric Intuition

- The graph of $y = mx + c$ is a **straight line**.
- Equal change in $x \Rightarrow$ equal change in y.
- $m > 0$: increasing; $m < 0$: decreasing; $m = 0$: constant.

Injectivity and Surjectivity

Injective

f is injective if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$.

$$mx_1 + c = mx_2 + c \implies m(x_1 - x_2) = 0.$$

So f is injective iff $m \neq 0$.

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Surjective

For $f : \mathbb{R} \to \mathbb{R}$,

$$y = mx + c \Rightarrow x = \frac{y - c}{m}.$$

All $y \in \mathbb{R}$ have preimages iff $m \neq 0$.

Inverse Function

If $f(x) = mx + c$ with $m \neq 0$,

$$f^{-1}(x) = \frac{x - c}{m}.$$

Inverse Function

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Verification

$$f^{-1}(f(x)) = \frac{mx + c - c}{m} = x.$$

Thus f^{-1} and f are mirror images over $y = x$.

Composition Property

Composition

If $f(x) = m_1x + c_1, \quad g(x) = m_2x + c_2$, then

$$(f \circ g)(x) = m_1m_2x + (m_1c_2 + c_1).$$

Hence linear functions form a group under composition when $m \neq 0$.

Matrix Viewpoint

For $T(\mathbf{x}) = A\mathbf{x}$:

- Injective $\Leftrightarrow \ker(A) = \{0\} \Leftrightarrow \det(A) \neq 0$.
- Surjective $\Leftrightarrow \mathrm{Im}(A) = \mathbb{R}^m$.
- Bijective $\Leftrightarrow$ square A with $\det(A) \neq 0$.

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Example:

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad A^{-1} = \frac{1}{5}\begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}.$$

Common Misconception

Beware

$f(x) = mx + c$ is not a **linear transformation** unless $c = 0$.

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In linear algebra, linear maps preserve:

$$T(a\mathbf{x} + b\mathbf{y}) = aT(\mathbf{x}) + bT(\mathbf{y}).$$

An affine term $+c$ breaks this property.

Examples

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$f(x) = 3x - 5$: injective, surjective, and

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Example 2

$f(x) = 0x + 2 \Rightarrow f(x) = 2$. Not injective, not surjective.

Advanced Problems (I)

Algebraic / Functional

1. If $f(x) = 2x + 3$, $g(x) = ax + b$ and $f \circ g = g \circ f$, find a, b.
2. Find all m, c so that $f^{-1} = f$.
3. If $f(g(x)) = x$, prove f, g are inverses and determine coefficients.

Suggested Reading

- Seymour Lipschutz, *Schaum's Outline of Linear Algebra*.
- Stephen Abbott, *Understanding Analysis*.
- Erwin Kreyszig, *Advanced Engineering Mathematics*.
- Zill, *A First Course in Differential Equations with Linear Algebra*.

Thank You!

Questions or discussions?